

Program overview

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Year 2022/2023
Organization Civil Engineering and Geosciences
Education Master Applied Earth Sciences

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
AES Master 2022 (New Master Programme)			
AES Faculty Core Module (MUDE)			
CEGM1000	Modelling, Uncertainty and Data for Engineers	12	
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AESM1001	Earth System, Natural Resources and Climate	7	
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AESM2001	Physical Principles of Earth System Observation	5,5	
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AES Faculty Core Module (MUDE)

CEGM1000	Modelling, Uncertainty and Data for Engineers	12
Responsible Instructor	Dr.ir. A.A. Verhagen	
Module Manager	Dr.ir. A.A. Verhagen	
Instructor	Dr.ir. W. Daamen	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr. A. Amiri Simkoeei	
Instructor	F.P. van der Meer	
Instructor	Dr.ir. H.R. Schipper	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Tsouvalas	
Instructor	Dr.ir. O. Morales Napoles	
Instructor	Dr.ir. R.W. Hut	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr. M.F.S. Tissier	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	P.K. Krishnakumari	
Instructor	Dr. A. Cabboi	
Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	Dr. D.V. Voskov	
Instructor	Dr. H. Wang	
Instructor	Dr.ir. M. Nogal Macho	
Instructor	Dr.ir. R.E.F. Lindeboom	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	E. Ragno	
Instructor	Dr. R. Taormina	
Instructor	Dr. A. Cicirello	
Instructor	J.O. Colomes Gene	
Instructor	M. Rinaldi	
Instructor	A. Jagadeesh	
Instructor	Dr. P.M.N. Mares Nasarre	
Contact Hours / Week x/x/x/x	4HC, 4W, 4PJ	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	All students have common needs at the start of their programme to further develop knowledge and skills to meet the standards of the TU Delft and its research- and design-oriented MSc programmes. The MUDE is one integrated module in which topics related to data, modelling and uncertainty quantification are applied to real engineering challenges. This module comprises two interlinked parts. The Theory, Application and Coding (TAC) part focuses on teaching and applying the fundamental concepts on modelling, uncertainty and data (MUD), as well as coding skills. In the Q1 Project part students work in multidisciplinary teams on cases in the context of a smart society, where they will apply the knowledge and skills from the TAC. In the Q2 project, the students work at the interface areas where the three topics overlap, creating opportunities for more integrated applications and the ability to focus on fields of interest per programme (when needed) while satisfying the same set of learning objectives. A gradually increasing complexity and openness of inquiry will be applied.	
Study Goals	After successfully completing the MUDE a student General - is able to describe and formulate a research question (or alternatively, design requirements) given a set problem and select the appropriate methodology and tools - is able to present a fitting work plan to investigate a set of research questions or design requirements - is able to compose a technical document using appropriate academic language and citation with references - is able to work in a collaborative group environment effectively - is able to code according to basic coding standards (e.g., consistency, readability, conciseness, structure, etc.) and collaborate with their peers via distributed control software (e.g., git) - can perform spatial / temporal / multivariate analysis of data to extract knowledge via physics-based modelling tools, data-driven approaches, and uncertainty quantification methods - can present and communicate with peers in Civil Engineering and Geosciences, results of analyses using specific modelling, uncertainty and data approaches with appropriate metrics and visualization techniques Modeling - can design a modelling framework (from problem conceptualization to governing equation setup) for a physical/engineering process - can translate the modelling framework into discretized equations and computer code - can mathematically formulate and solve an optimization problem and discuss its properties - can assess optimization and simulation models performance using a set of indicators Uncertainty - can derive relevant models and complete probability and statistic calculations for faculty-wide applications - can construct relevant models for probabilistic dependence, for example using bivariate distributions or discrete Bayesian	

	Networks - use deterministic models with probabilistic inputs to evaluate engineering questions to quantify model output uncertainty (in time and/or space) - use risk and reliability analyses concepts to describe systems, their characteristics and behaviour in time and space to support decision making under uncertainty Data - can identify parameters of interest, describe the functioning of common field/remote sensors used for measuring them, and explain resolution implications in space and time - is able to apply parameter estimation using observations and perform a quality assessment - can illustrate the procedure of gathering data from different sources, pre-processing it for analysis purposes, and sharing it according to FAIR principles
Education Method	Each week starts with a plenary session to engage students by introducing the topic and showing the relevance. There will be three blocks per week to work on Theory, Application and Coding. In each block, students will have to prepare themselves by watching videos / reading some documents / doing exercises. Support will be available. Next follows an interactive class session, where teachers and teaching assistants are available to discuss and review the materials, and to work on assignments. In addition, students spend one block on the project assignment.
Assessment	Formative assessment based on Jupyter Notebooks, in-class exercises and project deliverables. Summative assessment: - Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.
Expected prior Knowledge	University units, or equivalent, in Calculus, Linear Algebra, Probability & Statistics, Numerical Analysis, Python programming
Academic Skills	Analysis, coding, communication, visualization, project management, team skills
Literature & Study Materials	See Brightspace
Judgement	Formative assessment based on Jupyter Notebooks, in-class exercises and project deliverables. Summative assessment: - Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.
Permitted Materials during Exam	Pocket calculator, formula sheet
Collegerama	No

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AES Core Modules

AESM1001	Earth System, Natural Resources and Climate	7
Responsible Instructor	Drs. J.C. Blom	
Module Manager	Drs. J.C. Blom	
Instructor	Dr. J.E.A. Storms	
Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Prof.dr.ir. B. Taebi	
Instructor	Dr. G. Bertotti	
Instructor	Dr. M.W.N. Buxton	
Instructor	Dr. M. Vizcaino	
Instructor	Dr. H.A. Abels	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	In this module, students will first be introduced to the Earth System and human interventions in the Earth System, including the exploitation of natural resources and greenhouse-gas emissions that caused climate change, and the societal challenge of moving towards a carbon-neutral world. Through three narratives, students will study through which processes and on what temporal and spatial scales the different Earth system components interact. The first story line is named Solid Earth & Resources and looks at the deep level dynamic processes that make the Earth a unique body in the solar system. These processes will be outlined in terms of the plate tectonic theory which provides a unified framework for the evolution of the solid Earth. We will examine how these processes have evolved through time and how they have been responsible for the distribution of the continents and the formation of mountain belts, volcanoes and the evolution of our natural resources. The second story line is named the Climate System, in which students will gain a basic understanding of the Earth's energy budget and the natural and anthropogenic influences on past (paleo-)climates and our current and future climate. They will be familiarized with the carbon cycle and study the role of the global atmospheric and oceanic circulation in setting climate zones and weather. The third story line is named Source to Sink and starts at mountains and follows the pathway of water and sediment towards the river, delta and coastal-oceanic basins. Along the way we will investigate formation of sediment, the water cycle, formation of stratigraphy, vegetation and land use changes, and how all this is affected by the past and current climate and weather. Finally, students will appraise and reflect on the societal and ethical implications of past or future human interventions related to resources and the climate of the Earth System.	
Study Goals	After completing this module, students will be able to: Analyse the different components of the Earth system and the processes and time and spatial scales on which the different components of the Earth system interact. [ILO A] Examine the processes that generate and deplete the availability of natural resources. [ILO A, D] Distinguish the processes that play a role in the Earth's energy (im)balance and calculate their impact on climate. [ILO A, C] Identify the processes that underlie impacts of anthropogenic activities on the Earth System. [ILO A, C] Reflect on social and ethical implications of human interventions in the Earth System. [ILO F]	
Education Method	Lectures and group practicals/exercises, feedback via quizzes (formative) and intermediate assignment results (formative).	
Assessment	Written exam, based on integrated understanding, and mixed group assignments (e.g., as a debate/poster/presentation) . Group assignments (30%): LOs 3, 5 Individual written exam (70%): LOs 1, 2, 3 and 4 Note: ethical reflection as part of assignment report (and optionally in exam as well)	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	none, although basic knowledge in geology will be useful	
Literature & Study Materials	Collaboration with peers, Integration of knowledge, reflection on ethics, report writing. Provided course material, published online lectures and material, book/e-books	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	none	

AESM1002	Dynamics of Solids and Fluids	5.5
Responsible Instructor	Dr. H. Hajibeygi	
Module Manager	Dr. H. Hajibeygi	
Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	A. Barnhoorn	
Instructor	Dr. S.R. de Roode	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. A.M.H. Pluymakers	
Instructor	Dr. M.M. Boon	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides students with an understanding of basic concepts and mathematical solutions of fluid and solid dynamics with applications to atmosphere, oceans and the Solid Earth, preparing them for in-depth study of processes in specific Earth components. Students will learn how to derive the governing equations of fluid dynamics and solid dynamics, to explain how these are applied and simplified for flow and solid dynamics in the atmosphere, ocean and the solid Earth. Students will also carry out exercises as group assignments using python notebooks and model codes that exemplify simple models built on these governing equations. The topics covered are: Fluid dynamics: Governing Physical Equations Conservation of Momentum: equations of motion in an inertial frame and a rotating frame, Cartesian and spherical coordinates, advection, Eulerian and Lagrangian derivatives, vorticity equation, vorticity and circulation theorems, vorticity equation in a rotating frame Dimensional analysis and dimension-less numbers (Reynolds, Rossby, Prandtl, Froude, Peclet) Conservation of Mass: compressible and incompressible flows Equation of state, thermodynamic equations for ideal gases and for liquids and mixtures (first and second law of thermodynamics, Daltons law, phase behaviour). Buoyancy and thermal (static) stability Laminar versus turbulent flows and diffusion (as proxy of turbulence) Fluid dynamics: Applications to ocean and atmosphere Examples of flow in the atmosphere and oceans: geostrophic flows, convection, turbulence Boussinesq and anelastic approximation, vertical (pressure) coordinates, Hydrostatic balance. Primitive equations as basis of atmosphere and ocean circulation models. Fluid dynamics: Application to Flow in Porous Media Single-phase flow in porous media (Porosity, Permeability, Darcys law, Gov eq., solving potential, pressure & velocity) Single-phase heat-mass transfer in porous media (conduction and convection) Upscaling (heterogeneities, anisotropic) Diffusion/dispersion in porous media Introduction to multiphase flow in porous media (gov. equations, basics of kr & pc); Rock-fluid interaction. Solid Dynamics Solid Mechanics: elasticity, material strength, brittle to ductile deformation, effect of conditions (temperature, pressure, material properties). Applications to rock deformation in shallow and deep Earth, and ice flow.	
Study Goals	After completing this module, students will be able to: Distinguish flows and deformation in different parts of the Earth system (atmosphere, ocean and solid Earth). [ILO C] Perform dimensional analysis of flows in different media and explain appropriate approximations to the equation of motion. [ILO A] Characterise and derive the physical equations that underlie advection, diffusion, convection, conduction, elasticity, brittle and ductile deformation, and consider their application in models of Earth System components. [ILO A, C] Apply the physical equations to examples of flow and deformation in different media using python notebooks. [ILO C, D] Analyse and compare the behaviour of flow and deformation phenomena in the Earth System using simple python models. [ILO A]	

Education Method	Lectures and group practicals/exercises Feedback via quizzes (formative) and intermediate assignment results (formative).
Assessment	Assignments (50%) and a final exam (50%) Written exam is individual, Assignment is in a group.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Bachelor of Science in Engineering and Sciences, with basic knowledge of mathematics and physics.
Literature & Study Materials	1- Critical and algorithmic thinking 2- Design an algorithm 3- Writing scientific report 4- Presentation skills
Judgement	A mix of book chapters and lecture notes.
Permitted Materials during Exam	Reports and Exam.
Collegerama	NONE.
	No

AESM2001	Physical Principles of Earth System Observation	5.5
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Module Manager	Dr.ir. F. Lopez Dekker	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Dr. S.C. Steele-Dunne	
Instructor	Dr. C. Weemstra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Measuring is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Electromagnetic, seismic and gravity potential-field observations inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. The measurements can be acquired from spaceborne, airborne, surface and sub-surface-based sensors. This programme core module aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain geophysical variables. For example, students will learn how electromagnetic theory allows to use the intensity of radar echoes to yield information about rain rate, soil moisture, ocean roughness, or the layering of the subsurface. Similarly, they will learn how potential field theory can be applied to quantify mass changes of, e.g., the ice sheets. Students will be able to weigh the advantages and disadvantages of, for example, microwave versus visible and near-infrared observations for monitoring the Earth surface, and seismic and electromagnetic imaging in mapping the subsurface. Review basics of electromagnetic (EM) waves (propagation, polarization, spectra, Doppler, etc.) and propagation effects (attenuation, refraction, dispersion, polarimetric effects) Optical-infrared scattering and propagation (different reflection types, albedo, effect of wavelength, BRDF, Radiative transfer modelling of optical EM waves). Physical principles and limitations of VNIR/SWIR sensing systems (photographic systems, LiDAR, spectral systems including TIR) in terms of spatial and spectral resolution, sensitivity, atmospheric disturbances and corrections. Microwave scattering and propagation (Rayleigh, Mie scattering, rough surface, volume scattering, clear air, radiative transfer modelling of microwave scattering in the atmosphere and at the land surface). Radiometry (Radiometric Quantities and Units, Blackbody Radiation, emissivity at different frequencies). Radiative transfer modelling in passive microwave remote sensing of the Earths atmosphere and surface. Physical principles and limitations of microwave sensing systems (radiometer, non-imaging radar, imaging radar, ground-penetrating radar) in terms of spatial and spectral resolution, sensitivity, atmospheric propagation effects and corrections, radar range equation Seismic (including acoustic) waves (Newtons second law, stress tensor, equation of motion, elasto-dynamic wave equation, Plane P- and S-waves, spatial and temporal resolution) Potential fields (gravity, magnetic and electric fields) and their relationship to physical properties of a medium (Gauss law and its applications, Newtonian/Magnetic Potential, Laplace equation and harmonic functions, representation of potential fields, spherical harmonics) Choose an approach for the observation and measurement of a geophysical variable.	
Study Goals	After completing this module, students will be able to: Characterize electromagnetic waves and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and & subsurface). [ILO A] Describe the interaction of optical-infrared waves with the Earths atmosphere and surface and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of optical/infrared remote sensing in determining geophysical variables. [ILO A, D] Describe microwave scattering in the Earths atmosphere and at the Earths surface, and analyse the fundamental sensitivity to geophysical variables of interest. [ILO A] Describe the emission of thermal waves and microwaves from the Earths atmosphere and surface, and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of microwave remote sensing in determining geophysical variables. [ILO A, D] Explain mathematically how the elastodynamic wave equation is derived, characterize elastic waves, and explain their sensitivity to contrast parameters. [ILO A] Establish the linearized relationship between an observable and a given representation of the potential field, and synthesize the potential from functionals given on the surface of a sphere. [ILO A] Critically discuss the potential value and limitations of different types of measurements (I.e., different bands of the EM spectrum and EMR/seismic/potential field) and propose and defend a type of measurement to observe certain geophysical variables. [ILO F, G]	
Education Method	Lectures and group practicals/exercises. Feedback via quizzes (formative) and intermediate assignment results (formative).	
Assessment	Module is summatively assessed with a project, using 1 rubric (similar as thesis project) at the end of the period based on:	
	Assignment report (may comprise different parts, i.e., based on a several assignments, report includes reflection on LOs)	
	Presentation	
	Oral discussion (may include questions on LOs not directly covered in the assignment work).	

Expected prior Knowledge Literature & Study Materials Judgement	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Some background in electromagnetic theory and fundamentals of wave propagation. Solving linear partial and ordinary differential equations. Provided course material, published online lectures and material, book/e-books For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
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Year	2022/2023
Organization	Civil Engineering and Geosciences
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AES Discipline Core Modules

AESM3001	Atmosphere Dynamics and Air-Sea Coupling	6
Responsible Instructor	Dr. A.A. Nuijens	
Module Manager	Dr. A.A. Nuijens	
Instructor	Prof.dr. J.D. Pietrzak	
Instructor	Prof.dr. C.A. Katsman	
Contact Hours / Week x/x/x/x	0/0/16/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This discipline core will provide students with the deeper physical understanding to comprehend and study the complex atmospheric flows that define climate and weather. The module will use interactive lectures to introduce advanced dynamical principles to explain global climate, the planetary circulation, weather systems in the tropics and extratropics and coupled atmosphere-ocean phenomena that set weather and climate variability. In these lectures, laboratory experiments with a rotating tank or visualizations of atmospheric flows are a starting point to describe the complex flows in our atmosphere. Each week, the students will be guided during practical lab work/assignments in which material collected during the tank experiments, toy models or observational datasets are used and analyzed to describe uncoupled and coupled phenomena in the atmosphere.	
Study Goals	After completing this module, students will be able to: 1. Analyse data of the horizontal and vertical structure of atmospheric pressure, temperature, humidity and wind to explain the general circulation of the atmosphere and climate zones [ILO A,B] 2. Apply conservation of heat, moisture and momentum to explain the origin and features of tropical and extratropical weather phenomena [ILO A] 3. Assess the role of air-sea coupling on atmospheric dynamics on short (weather) and long (climate) timescales [ILO A] 4. Analyse and evaluate the sensitivity of circulations and climate by running experiments with idealized simulations and toy models [ILO A,B,C] 5. Design a hypothetical lab experiment to study the influence of climate change on atmosphere dynamics [ILO A,C,E] 6. Discuss processes that cause uncertainty in weather and climate prediction [ILO C,F,G] (ILO = Intended Learning Outcomes of the AES programme)	
Education Method	Plenary discussions and quizzes during lectures and labs, practical (lab) assignments to be handed in, midterm exam.	
Assessment	Written theory exam.	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	Faculty programme core, AES programme core.	
Literature & Study Materials	Demonstrating conceptual understanding through the application of theory to new situations using relevant equations. Extracting, evaluating and discussing modeling output and observational data. Creating new experiments.	
Judgement	Textbook: Marshall, Plumb: Atmosphere, Ocean and Climate Dynamics: An Introductory Text. Lecture notes provided with slides.	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	Lecture notes, personal notes and calculator.	
	No	

AESM3002	Earth Observation Technologies	6
Responsible Instructor	Dr.ir. D.C. Slobbe	
Module Manager	Dr.ir. D.C. Slobbe	
Instructor	Prof.dr.ir. R.F. Hanssen	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkoei	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	The goal of this module is to familiarize students with the process of specifying and designing an observational mission/campaign (e.g., a new satellite mission or a ground measurement campaign). This process includes the interpretation and analysis of user needs and their translation to observational requirements, the high-level design of possible technical solutions, and the evaluation of the expected observational performance with respect to user requirements. The module will adopt an inquiry-based learning strategy, where groups of 4-5 students will work throughout the module on an observational mission/campaign motivated by topical scientific or societal issues. Each group of students will be mentored by a domain expert to guide their exploration of available and emerging EO technologies, potential trade-offs in mission/campaign design, the availability of suitable forward and inverse models, etc. The required theoretical background will be provided through a combination of classroom lectures and digital learning. This allows the students to acquire a broad foundation and deepen their knowledge on topics most relevant to their observational mission/campaign. Students will begin by formulating, consolidating, and prioritizing user requirements and translating those to mission/campaign requirements. Students will draw on their prior/existing understanding of the underlying physical principles to select a suitable observation technique to observe the variable(s) of interest. Students will develop a preliminary design for an observational mission/campaign that reconciles user requirements with the capabilities of the selected techniques. Where appropriate, students will be provided with real or synthetic observation data that they will analyse to characterize the observations obtained from their mission/campaign, perform a quality assessment, estimate the parameters of interest and reflect on the ability of the mission/campaign to meet the user requirements. Mission/campaign proposals will be presented in a written report (proposal) and orally (pitch).	
Study Goals	After completing this module, students will be able to: -Formulate user requirements that can be assessed based on parameter estimation [ILO A,B,E] -Evaluate the principles and limitations of generic classes of observation techniques, observation platforms, and data processing techniques [ILO A,B] -Analyse third-party user requirements and translate these into system requirements of an observational system (mathematical and physical) [ILO D] -Design an observational mission to gather the requested observations [ILO A,B,D,E,F,H] -Analyse results from an observation mission by estimating the parameters of interest, including quality assessment [ILO B] -Reflect on the analysis results in relation to third-party stakeholders [ILO E,F] -Contribute to effective group work and communicate orally and in written form on the project results at an academic level [ILO G,I]	
Education Method	Lectures and group practicals/exercises, Quizzes Peer and expert feedback during project sessions.	
Assessment	Module is summatively assessed with a group project, using 1 rubric (similar as thesis project) at the end of the period based on: Assignment report (report includes reflection on LOs) Presentation Oral discussion (may include questions on LOs not directly covered in the assignment work). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Faculty and program core modules	
Academic Skills	-Research skills -Data analysis -Written and verbal communication -Listening -Reasoning -Critical thinking	
Literature & Study Materials	Lecture slides and selected papers	
Judgement	Judgement will be based on rubric (to be developed) For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

AESM3003	Geo-Energy Engineering Applications	6
Responsible Instructor	Dr. A. Barnhoorn	
Module Manager	Dr. A. Barnhoorn	
Instructor	Dr. H. Hajibeygi	
Instructor	A. Daniilidis	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module will provide a general overview of the application areas of geothermal energy, petroleum exploration and production, and energy storage in the subsurface. The students will learn in this core module how these geo-energy application areas contribute to the energy demand of the world, in what way they can contribute to the transition towards a carbon-neutral world, what the opportunities, boundary conditions and consequences are of these applications. The students will learn the basic techniques and principles of reservoir characterization and single phase flow.	
	This core module consists of the following components:	
	Basic principles of reservoir characterization and of single phase flow and how these two complement/interact with each other	
	Introduction into energy transition towards a CO2 neutral future energy production, what is the role of the subsurface. Demand, potential and requirements	
	Basic principles governing and applications of Geothermal Energy (geothermal potential, flow and heat in the subsurface), include role of reservoir characterization and flow	
	Basic principles governing and applications of Subsurface Storage (storage of CO2, energy carriers, boundaries of safe operations, cyclic storage), include role of reservoir characterization and flow	
	Basic principles governing and applications of Petroleum Exploration and Productions, include role of reservoir characterization and flow	
	Basic principles governing and applications of the Effects of Subsurface Engineering (Subsidence, seismicity, changes to reservoir)	
	The course will be a mixture of lectures and assignments to determine the role of the three applications for energy production and how they play a role in for filling demand for energy towards the future. The effect of those operations are included.	
	By doing those individual assignments a conceptual plan will be build that provides the students with an integrated approach to the use of the subsurface for the world's future energy demand.	
Study Goals	After completing this module, students will be able to:	
	Explain what the field of Geo-Energy entails, how the subsurface can be utilised and what its role is in the energy transition. [ILO A]	
	Analyse the basic concepts of single-phase flow and reservoir characterization in relation to geothermal energy, subsurface storage, petroleum exploration and production applications and the effects of engineering in this subsurface [ILO A, C]	
	Evaluate how single-phase flow and subsurface characterisation interact at different temporal and spatial scales [ILO A]	
	Develop a conceptual plan that compares and integrates the different applications utilizing the subsurface for the energy demand towards the future. Quantify the components of the conceptual plan using the physic principles that describe subsurface phenomena. Test this problem towards the sensitivity of different components/issues (flow, mechanics, heterogeneity) [ILO C, D, E, F, H, I]	
	Use written and oral communication skills to effectively exchange results and opinions with researchers, engineers, and Applied Earth Sciences stakeholders. [ILO G, H]	
Education Method	Lectures, assignments. Feedback on intermediate assignment results.	
Assessment	Group assignment (100%)	
	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Faculty programme core, AES programme core	
Academic Skills	Academic skills:	
	- group work	
	- effective oral communication	
	- scientific writing	
	- critical thinking, development of concepts	
	- integration of data and disciplines	
	- assess impact and uncertainty	
Literature & Study Materials	All material (handouts, readers etc.) and links to online material will be provided during the course via Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
	Group assignment (100%)	
Permitted Materials during Exam	n.a.	

AESM3004	Economic and Structural Geology	6
Responsible Instructor	Dr. M.W.N. Buxton	
Module Manager	Dr. M.W.N. Buxton	
Instructor	Dr. G. Bertotti	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This module will equip students with the theory and knowledge to understand the nature, origin and factors controlling primary solid/mineral raw materials in order to understand the physical and chemical characteristics that influence and control the behaviour in terms of recovery of value and utilisation. It consists of two units: 1. Economic Geology in which the characteristics of different ore body styles of mineralisation, the origin, controls, formation mechanisms and abundance of primary and secondary raw materials are explained. 2. Geomechanics and structural geology where the role of structural geology and its significance for the formation of ore bodies are presented.	
Study Goals	After completing this module, students will be able to: Classify the variety of mineral resource commodities of primary/secondary origin and discuss the relevance to society. [ILO A] Illustrate the spatial and temporal distribution of mineral resources in the Earth crust by means of specific examples (Precious metals, base metals, ferrous metals, non-metallics, secondaries). [ILO A] Compare evolutionary concepts about the origin of mineral deposits by means of specific examples (Precious metals, base metals, ferrous metals, non-metallics, secondaries). [ILO A,C] Classify different mineral deposit styles based on geological features. [ILO A,C] Relate different processes that give rise to different ore deposit styles to their genetic classification based on the observed geological features. [ILO A,C] Define rock stress and strength and how rocks respond to different styles of deformation. [ILO A] Reconstruct the structural evolution of multiscale rock bodies on the basis of documents such as stress-strain diagrams, seismic sections and outcrop data. [ILO A,C] Appraise the significant processes that lead to formation of an ore deposit during structural deformation. [ILO A,C] Work effectively in a team to define, plan and execute a project assignment and to report the outcomes by means of oral presentation and written report. [ILO G,H,I]	
Assessment	Integrative report on assignment at module-level (50% of module grade), written exam unit 1 (25%, 1-dec, 5.8, compulsory), report on assignment unit 2 (25%, 1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Faculty programme core, AES programme core.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES A-Modules

AESM301A	Atmospheric Processes and Modelling	9
Responsible Instructor	Dr. A.A. Nuijens	
Module Manager	Dr. A.A. Nuijens	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. A.P. Siebesma	
Instructor	Dr. S.R. de Roode	
Instructor	Dr. M. Vizcaino	
Instructor	Dr.ir. B.J.H. van de Wiel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This module is aimed at students with an interest in the field of applied meteorology such as wind and solar energy prediction or climate impacts on land use), and students interested in the design, analysis and assessment of weather and climate models.	
Study Goals	The main objective of the module is to provide students with a solid understanding of atmospheric processes that play a key role in climate and that have a strong impact on daily life: winds, clouds and radiation, storms and precipitation. This module will explain students how these small-scale processes are coupled to large scale circulations and why their formulation in general circulation and climate models is particularly important for climate prediction.	
	The module comprises two units that focus on deepening process knowledge and analytical thinking: Land-atmosphere interactions and Water in the atmosphere. These will cover atmospheric processes strongly linked to turbulence and radiative transfer across all scales that are not explicitly represented in general circulation models. The third unit called Climate Modelling provides students with hands-on experience in the use and application of state-of-the-art general circulation models, including an assessment of the uncertainties in current climate projections.	
Education Method	After completing this module, students will be able to:	
	1. Explain the physics and dynamics of the transport of energy, water and momentum between the surface, the atmospheric boundary layer and the broader atmospheric circulation [ILO A] 2. Apply simplified models of turbulence and energy and water exchange, including convection and clouds [ILO B,C] 3. Reflect on the parameterization and coupling of atmospheric processes in general circulation models and their role in current uncertainties in climate prediction [ILO C,F,G] 4. Analyse climate simulations for the purpose of process understanding as well as assessment of climate change impacts and mitigation/adaptation policies [ILO A,B,C,E,F] 5. Assess the influence of land-atmosphere coupling, turbulence, convection and clouds on large-scale circulations (weather) and climate. [ILO A,C,F] 6. Hypothesize how land-atmosphere coupling, turbulence and convection and clouds are influenced by changes in weather and climate [ILO A,C,F] (ILO = Intended Learning Outcomes of the AES programme)	
Assessment	Unit Water in the Atmosphere: - 2 hours of classical lectures and 2 hours of supervised computer-based and regular exercises. Unit Land-Atmosphere Interaction: - 2 hours of lectures including short mini-exercises, table-top experimental demonstrations, article discussion in class, 2 hours of supervised theoretical exercises. Unit Climate Modelling: - Lectures, exercises (including paper analysis/discussion), guided assignments See also Brightspace for updated and more information of this module's units.	
	Summative assessment: Two written exams on unit level (25% each), 1 assignment on unit level (25%) and one module level synthesis assignment (25%). Formative assessment: Unit Water in the Atmosphere: - Computer-based and regular (homework) exercises are supervised with ample feedback. Unit Land-Atmosphere Interactions: - Exercises made under supervision of the instructors, feedback to the experimental work by the students (both on performance and on reporting) and a midterm exam. Unit Climate Modelling: Exercises (including paper analysis/discussion) and assignments A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Combination with DC-CW (Atmosphere Dynamics and Air-Sea Coupling).	
Academic Skills	Deepening scientific knowledge, analytical and critical thinking, problem solving, synthesizing information and scientific writing.	
Literature & Study Materials	Unit Water in the Atmosphere: - "Atmospheric Science, An Introductory Survey", by John Wallace and Peter Hobbs	
	Unit Land-Atmosphere interaction: - The book: Transport in the Atmosphere-Vegetation-Soil Continuum, by Moene and Van Dam, Cambridge University Press. Additional background material and/or papers will be provided via BrightSpace. Experimental material is provided in the class.	

	Unit Climate Modelling: - Primary Course Book: Climate Modeling Primer (second course book and Practice) by McGuffie and Henderson-Sellers. Secondary Course Book: Climate System Modeling, by Kevin Trenberth - IPCC reports (Practice) https://www.ipcc.ch/reports/ See Brightspace for more detailed information on literature and study materials for the module and units.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Calculator. See Brightspace for updated information on permitted material during the exam.
Collegerama	No

AESM302A	Geo-Data Analysis and Geodesy	9
Responsible Instructor	Dr.ir. A.A. Verhagen	
Module Manager	Dr.ir. A.A. Verhagen	
Instructor	Dr.ir. H. van der Marel	
Instructor	Prof.dr.ir. P.J.G. Teunissen	
Instructor	Dr. R.E.M. Riva	
Instructor	Dr. P.G. Ditmar	
Instructor	A. Barnhoorn	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Contents	This module targets students interested in learning how to rigorously use geo-data to estimate and monitor changes in the shape of the Earth's surface and its gravity field. The signals of interests can be related to local human activities, such as gas or ground-water extraction, or, for example, related to climate change, such as ice-mass losses in Greenland or Antarctica. In this data-oriented module, students will acquire the skills and theoretical background required to process Earth observation data in order to retrieve the signals of interest, in particular by using Fourier analysis methods. A key aspect of this module is dealing with errors in the data and assessing the quality of the input data as well the estimated parameters. Students will learn how to characterize different types of noise, and to make the best possible estimate of the parameters of interest in presence of noise. By learning about hypothesis testing, students will be able to select the model that best fits a set of measurements. Students will also learn to link the estimated parameters (for example, surface deformation rates) to the underlying geodynamical processes. For this purpose, students will learn about some key geophysical processes, such as Earth tides, glacial isostatic adjustment, and tectonics processes, including Earthquakes. Students will follow this module in combination with Earth Observation Technologies, the Earth Observation discipline core. The module includes three units: Statistical geo-data analysis. Signal Processing. Geodesy and Geodynamics	
Study Goals	After completing this module, students will be able to: Assess the quality of EO data and derived products Design and apply hypothesis testing procedures to select the model which best represents physical reality Apply spectral analysis techniques to extract relevant geophysical information from EO data Apply geodetic observation and analysis techniques to quantify and characterize changes in the shape of the Earth and its gravity field Analyse the link between geodetic observables and the underlying geodynamical processes Present findings in a precise and organized way, both numerically as well as graphically	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three assignments on unit level (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory).	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	Combination with AESM3002	
Literature & Study Materials	Data Analysis, Visualization	
Judgement	See Brightspace for more detailed information on literature and study materials for the module and units.	
	Three assignments on unit level (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory).	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	n.a.	

AESM303A	Geo-Data and Geo-Informatics	9
Responsible Instructor	Dr. R.C. Lindenbergh	
Module Manager	Dr. R.C. Lindenbergh	
Instructor	Prof.dr.ing. R. Klees	
Instructor	Dr.ir. H. van der Marel	
Instructor	Prof.dr.ir. P.J.G. Teunissen	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This is the module for students interested in exploring, mining and communicating the wealth of relevant information in state-of-the-art geospatial data. Different ways to visualize and process geospatial data, in different formats and projections, on geographic information systems will be explored. It will be discussed how to assess the quality of input Earth observation data, and how this quality propagates through a processing chain towards a quality description of a final product. Methodology will be analysed to assess the spatial-temporal contents of data in terms of repetitive patterns and the scales at which information is present in both the spatial and temporal domain. Finally, it will be discussed how such different information can be extracted from data, and how the significance of the extracted information can be accessed and communicated to different stakeholders in effective and attractive ways.	
Study Goals	After completing this module, students will be able to: Assess the quality of EO data and derived products [ILO B] Design and apply hypothesis testing procedures to select the model which best represents physical reality [ILO A,B] Apply spectral analysis techniques to extract relevant geophysical information from EO data [ILO B] Model and estimate the spatial and temporal variability of EO data and relate it to the underlying geophysical processes [ILO B,C] Select and apply appropriate geo-informatics tools to extract, process and communicate information from EO data [ILO B,D] Present findings in a precise and organized way, both numerically as well as graphically [ILO G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three unit level assignments (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory).	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5.8 per Exam/Assessment.	
Academic Skills	Combination with DC-EO. Written and verbal communication. Reasoning. Attention to detail. Data analysis.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Simple, non-graphical calculator	
Collegerama	No	

AESM304A	Flow and Simulation of Subsurface Processes	9
Responsible Instructor	Dr. D.V. Voskov	
Module Manager	Dr. D.V. Voskov	
Instructor	Dr. J.E.A. Storms	
Instructor	Prof.dr. P.L.J. Zitha	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Dr. H. Hajibeygi	
Instructor	Dr. P.B.R. Bruna	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	In this module, students acquire the necessary tools and knowledge to accurately model the building blocks of subsurface reservoirs and to accurately model flow of fluids/energy through these subsurface reservoirs. The 3 units focus on 1) analyse sedimentary and structural models of how the effective rock properties and the different facies, faults and fractures are distributed in subsurface reservoirs 2) describe, analyse and analytically model flow of fluids and energy through those reservoirs (both single-phase as two-phase fluids) and 3) describe, analyse and model flow numerically through these subsurface reservoirs. The students develop these models themselves, analyses the outcomes of these models and reflect on sedimentary and structural heterogeneities, variations, and uncertainties in these models. Particular attention will be paid to the role of scale of effective properties in these three approaches and how these units can be used to understand, solve and predict the key issues within geo-energy engineering applications such as geothermal energy, subsurface storage, hydrocarbon production and consequences of subsurface engineering.	
Study Goals	After completing this module, students will be able to: Evaluate how to build process and stochastic reservoir models and choose appropriate rock properties or facies distribution [ILO A,C] Analyse the physics and develop an analytical model for two-phase flow, thermal processes and poroelasticity through porous media with different assumptions and for different applications [ILO C] Design a stable and consistent numerical method for modelling of flow and transport in porous rocks [ILO C] Implement and create the three model types (reservoir model, analytical two-phase flow, numerical flow model) with geo-energy related software and codes [ILO A,D] Apply the models to geo-energy related test cases and analyse the solutions with respect to the role of uncertainties, sensitivities, relationships and consequences critically for the different test cases [ILO F] Work as a team on subject related problems and report findings and interpretations, including codes and choices made, in a structured and consistent way. [ILO G,I]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three unit level reports (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Combination with DC-GE.	
Academic Skills	Critical and algorithmic thinking Design an algorithm Writing scientific report	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	Unit reports based on assignments (75% of module grade) Synthesis assignment (25% of module grade) For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	NA	
Collegerama	No	

AESM305A	Characterisation of the Subsurface	9
Responsible Instructor	Dr. J.E.A. Storms	
Module Manager	Dr. J.E.A. Storms	
Instructor	Dr. G. Bertotti	
Instructor	Drs. J.C. Blom	
Instructor	Dr. P.B.R. Bruna	
Instructor	Dr. H.A. Abels	
Instructor	Dr. A.M.H. Pluymakers	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module focusses on the structural and sedimentological architecture of the rocks in the Earth that are important for Geo-Energy Engineering applications. The properties of rocks, as well as their variabilities on all scales, are assessed and quantified. Knowledge of the various processes taking place in the Earth and at the Earth's surface help the students assess and predict subsurface architecture and properties. The units consists of an understanding and modelling of how subsurface reservoirs are build up with respect to reservoir properties and facies distributions. How those changes in properties and distributions can be correlated to variations in sedimentary deposition due to past climate fluctuations within different tectonic settings. How those changes can be complemented within the structural framework in the subsurface (tectonics, deformation, compaction, faulting and folding). Particular attention will be placed to heterogeneity, variations and the issue of scale in subsurface characterization. Furthermore, the students will be taught how process-based understanding of sedimentary and tectonic processes is used to accurately populate subsurface reservoir models and how to accurately predict variations in these properties (both in space and in time).	
Study Goals	After completing this module, students will be able to: Evaluate and quantify process controlling spatial and temporal changes in sedimentological and structural properties of sedimentary successions. [ILO A,C] Evaluate and quantify state of stress in sedimentary basins, the way rocks deform and the mechanic factors controlling their response to geo-energy activities. Predict spatial and temporal changes of stress, strain and mechanic properties in the subsurface. [ILO A,C] Interpret the various sedimentary successions and structural features from theory, field, seismic and borehole data. [ILO A,C] Evaluate how to build sedimentary and structural reservoir models and choose appropriate rock properties and property distributions to characterise the model (including sources of uncertainties) [ILO A] Analyse the origin and scales of heterogeneities in sedimentary deposits and structural features and integrate these into representative elementary volumes [ILO A] Appraise the models in geo-energy related test cases and analyse the solution and investigate the role of uncertainties, sensitivities and relationships for the different test cases [ILO D,F] Work as a team on subject-related assignments and report findings and interpretations, including codes and choices made, in a structured and consistent way. [ILO G,I]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three unit level reports (25% each, 1-dec, 5,8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5,8, compulsory).	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Literature & Study Materials	Combination with DC-GE.	
Judgement	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESM306A	Extraction Processes and Consequences of Raw Materials	9
Responsible Instructor	Dr. M.W.N. Buxton	
Module Manager	Dr. M.W.N. Buxton	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr. G. Giardina	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. M. Soleymani Shishvan	
Instructor	Dr. J. Gebert	
Instructor	Ir. M. Keersemaeker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module contains three interrelated units namely Extraction Methodologies, Residual Materials from Post Extraction Processing and the Impact of Primary and Secondary Mineral Raw Materials on the environment.	
Study Goals	After completing this module, students will be able to: Explain the environmental and societal implications associated with the mining related processes to enable the development of innovative management strategies for geo-resources. [ILO C,D] Explain the different extraction methodologies and consequences and implications associated with them. [ILO D] Evaluate the nature of the residual materials from post extraction processing specifically the origin, handling, stacking and storage of mining residues. [ILO B,D] Explain the physical, chemical and fluid Interactions and dynamics for mining residues. [ILO C] Analyse and interpret appropriate geophysical and InSAR monitoring data to detect anomalies and consequences over different time scales of mining residues. [ILO B,C] Translate environmental challenges and economic opportunities in solid waste management into scientific research questions and/or engineering opportunities. [ILO B,C,D,F] Design an integrated plan for the extraction, waste storage and handling, targeting minimal waste and mitigation type strategies for risk/hazards associated with mine waste. [ILO D,E,F] Consider the UN, ICMM and EU goals and regulations when developing engineering solutions for waste management options. [ILO C,F] Work effectively in a team to define, plan and execute a project assignment and to report the outcomes structured and consistent by means of oral presentations and written report. [ILO G,H,I]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	One integrated summative assignment (design, case study) for all three units (1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Combination with DC-GR.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESM307A	Earth Deformation Processes Across Scales	9
Responsible Instructor	Dr. R.E.M. Riva	
Module Manager	Dr. R.E.M. Riva	
Instructor	Dr.ir. H. van der Marel	
Instructor	Prof.dr.ir. P.J.G. Teunissen	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	A. Barnhoorn	
Instructor	Dr. G. Bertotti	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module provides the knowledge and skills to understand, predict and characterise Earth deformation processes from continental (e.g., glacial isostatic adjustments and plate tectonics) towards reservoir scales (e.g., folding, faulting and compaction). Geodetic and geophysical observation techniques will be used to quantify these deformation processes, by extracting physical parameters and assessing their uncertainties. In addition, students will learn to relate the observed movements to subsurface engineering (e.g., resource extraction, storage, tunnelling) or natural processes (e.g., plate tectonics, earthquakes). The module contains three components, 1) Statistical geo-data analysis, 2) Geodesy and Geodynamics, and 3) Geomechanics and Structural Geology.	
Study Goals	After completing this module, students will be able to: Design and apply hypothesis testing procedures to select the model which best represents physical reality [ILO B] Apply geodetic observation and analysis techniques needed to quantify, characterize and explain changes in the shape of the Earth and its gravity field, and changes and expressions of crustal structures [ILO A,B,C] Evaluate the mechanical and deformation response and expressions of rocks to varying stresses within the shallow part of the Earth's crust [ILO B] Analyse the link between geodetic observables and the underlying geodynamical and geomechanical processes from reservoir to global scales, including the effects of subsurface engineering activities [ILO B, D] Present findings in a precise and organized way, both numerically as well as graphically [ILO G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three reports on unit level based on assignments (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory).	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5.8 per Exam/Assessment.	
Academic Skills	Combination with DC-EO or DC-GE.	
Literature & Study Materials	Critical thinking. Use of models and observations to solve practical problems. Awareness of issues related to the use of available observations for the purpose of explaining natural and anthropogenic processes within the solid earth.	
Judgement	See Brightspace for more detailed information on literature and study materials for the module and units.	
	Final grade consists of: three assignments on unit level (25% each) and one module level synthesis assignment (25%). Each component needs to be completed with a minimum grade of 5.0 (out of 10.0). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable.	
Collegerama	No	

AESM308A	Climate Modelling and Remote Sensing	9
Responsible Instructor	Dr. S.C. Steele-Dunne	
Module Manager	Dr. S.C. Steele-Dunne	
Instructor	Prof.dr.ir. R.F. Hanssen	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkooei	
Instructor	Dr.ir. D.C. Slobbe	
Instructor	Dr. M. Vizcaino	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module complements the Climate & Weather core module. It is aimed at students from the Climate & Weather learning line who are interested in the synergy offered by the combination of remote sensing and climate models to understand, predict and model the wide range of processes governing our present-day coupled climate system. The objective of this module is to provide students with a physical understanding of these processes and the skills needed to observe and assess those processes using ground-based and space-borne observations, and coupled global models. Ultimately, the student will be able to understand and analyse both one specific component of the climate system (e.g. cryosphere, terrestrial water cycle, ocean), and on a broader level, the interactions between them using a combination of observations and modelling. Students will learn how to formulate, consolidate, and prioritize their data needs, and will draw on their prior/existing understanding of the underlying physical principles to select a suitable observation technique to observe the variable(s) of interest. By developing a preliminary design for an observational mission/campaign they will become familiar with the trade-offs commonly made to reconcile requirements from a modelers perspective with the capabilities of the selected techniques. They will analyse real or synthetic observation data, perform a quality assessment, estimate the parameters of interest and reflect on the ability of the mission/campaign to meet their needs in terms of parameterizing processes of interest, constraining models and improving predictions. The module comprises two units that, together, offer the students a thorough understanding of how large-scale climate processes can be observed and modelled. Earth Observation Technologies (6 EC) consists of the core module of the Earth Observation learning line and familiarizes students with the process of specifying and designing an observational mission/campaign (e.g., a new satellite mission or a ground measurement campaign). This process includes the interpretation and analysis of user needs and their translation to observational requirements, the high-level design of possible technical solutions, and the evaluation of the expected observational performance with respect to user requirements. The unit Climate Modeling and Prediction (3 EC) from the Climate & Weather learning line is shared with other modules and provides students with hands-on experience in the use and application of state-of-the-art general circulation models, including an assessment of the uncertainties in current climate projections.	
Study Goals	After completing this module, students will be able to: Formulate, consolidate, and prioritize remote sensing data requirements to study processes in the coupled climate system and validate and inform their parametrization in climate models [ILO A,E,F] Select a suitable observation technique to observe the variable(s) of interest, drawing on their prior/existing understanding of the underlying physical principles [ILO A,E] Characterize real or synthetic observation data, perform a quality assessment, estimate the parameters of interest and reflect on the ability of the available data to meet their requirements. [ILO A,B,C,F,G] Construct (simplified) models of climate variables and evaluate these with remote sensing data. [ILO A,B,C,E] Analyse climate simulations for the purpose of process understanding (climate science) as well as for the interpretation of remote sensing data records. [ILO A,B,C] Reflect on how integration of remote sensing data in climate modelling may improve our understanding and prediction of climate variables. [ILO A,C,E,G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	unit 1: project unit level (67%, 1-dec, 5.8, compulsory); Unit 2: Assignment (25%, 1-dec, 5.8, compulsory); and one module level synthesis assignment (8%, 1-dec, 5.8, compulsory).	
	A successful completion of the module requires a minimum grade of 5.8 per Exam/Assessment.	
Expected prior Knowledge	Combination with DC-CW.	
Academic Skills		
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	See Brightspace for more detailed information	
Colleggerama	No	

AESM309A	Climate Change and Dynamic Landforms	9
Responsible Instructor	Dr. M. Vizcaino	
Module Manager	Dr. M. Vizcaino	
Instructor	Dr.ir. H. van der Marel	
Instructor	Dr. J.E.A. Storms	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. R.C. Lindenberg	
Instructor	Dr. H.A. Abels	
Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module is aimed at students with an interest in the physics of climate change, its effects on the natural environment and how those can be analysed through geospatial data. In the climate modelling unit, student will learn about the numerical, computational and modelling concepts that underlie general circulation models and coupled climate models, which are the primary tools to explain and/or predict the dynamics of past and future climate. The natural environment is here represented by rivers and deltas, which provide an instructive and societal-relevant study case to learn about the interaction between climate change and the surface of the solid Earth. Finally, the Earth observation unit has a strong hands-on identity, which complements the geological characterization of landforms by teaching how to deliver remote sensing products that can be used in geophysical studies. This module will fit into a study-route that addresses how the natural environment is affected by past and future climate change at decadal to secular scales, by using numerical models of the driving physical processes as well as geological and instrumental observations.	
Study Goals	After completing this module, students will be able to: Analyse the interaction between climate change and the solid Earth's surface, with particular attention to the timescales involved. [ILO A,C] Explain the origin of present-day morphodynamic features by integrating geological and geospatial data. [ILO A,B] Analyse climate simulations for the purpose of understanding the driving physical processes. [ILO C] Analyse source-to-sink sedimentary systems and records in relation to tectonics and climate amongst other parameters. [ILO A] Extract geophysical parameters from geospatial data. [ILO B] Present findings in a precise and organized way, both numerically as well as graphically [ILO A,G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Unit 1-3 assignment on unit level (25% each, 1-dec, 5,8, compulsory) and one module level synthesis assignments (25%, 1-dec, 5,8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES B-modules

AESM401B	Climate and Weather B-module	15
Responsible Instructor	Dr. M.A. Schleiss	
Module Manager	Dr. M.A. Schleiss	
Instructor	C.M.H. Unal	
Instructor	Prof.dr. P.L.J. Zitha	
Instructor	Dr. R.E.M. Riva	
Instructor	Dr.ir. B. Wouters	
Instructor	Dr. M. Vizcaino	
Instructor	Dr. D.P. Donovan	
Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x/x	week 1: 12 PJ week 3, 5, 6 : 8 PJ week 7, 8: PJ	
Education Period	4	
Start Education	4	
Course Language	English	
Course Contents	The lab and the theory part of the B-module are connected and inquiry-based. Theory (8EC) Students can choose theory units depending on the choice for a Geohazards or Climate Impacts Lab. Time series analysis is compulsory. Two units are to be selected from subset 1, and one unit is to be selected from subset 2. Lab (4EC) Individual or groups of max. 4 students The lab has a central place in the module, triggering the inquiries and as the place where the students apply the theory to a challenge related to monitoring and/or prediction of climate impacts or geohazards. The student (team) defines a topic related to the challenge within a predefined theme that they will work on during the Lab. They collect analyse and interpret data to address or solve their posed topic. The lab includes an ethical reflection (0.5 EC) on the specific challenge and approach. Fieldwork (3EC) Group size: 3 - 5 During the fieldwork, students will work in teams to define an objective related to the understanding of a phenomenon/process in weather and climate relevant to a societal challenge. Students start with a project planning phase (0.5EC), in which they receive instruction, and they design a measurement experiment, which will be reviewed. In a second phase they will implement the measurements, collect data and process the data (1.5EC). Thereafter they will analyse and interpret the data, present their findings and provide recommendations for future work (1EC).	
Study Goals	General objectives for theory, labs and fieldwork: After completing the module students will be able to: - Identify open issues in the processing and/or interpretation of Earth system data records based on the outcomes of the lab project and design a development roadmap to address them. [ILO E,F] - Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab and Fieldwork projects in a clear and convincing manner, both orally and written. [ILO F,G,H] Theory/Lab (module specific): - Explain which techniques are needed to measure or model specific geophysical parameters governing the processes relevant to a societal challenge. [ILO A,B,C,E] - Evaluate the temporal and spatial requirements for measurements or models to monitor and predict these processes. [ILO A,B,C] - Combine and analyse data records to understand processes and their relationships on relevant scales [ILO A,B,C] - Distinguish the different sources of uncertainty in observational and in model data. [ILO A,B,C] - Analyse the skills and errors of a model by comparing with observations. [ILO A,B,C] - Evaluate the sensitivity of geophysical models to key parameters and boundary conditions. [ILO C] - Analyse, visualize and interpret and present findings in a clear and convincing manner. [ILO A,B,F,G] Fieldwork: - Plan and design a field campaign that is appropriate for the physical process to be measured. [ILO A,D,H,J] - Collect data in the field using different measurement techniques. [ILO A,J] - Explain and quantify the error sources associated with the field measurements. [ILO A,B] - Process and analyse the data collected in the field to give meaningful constraints on the physical process. [ILO B,C] - Effectively communicate with peers, assessors and clients. [ILO G,H,I] - Contribute to a project as a team player and to the overall project management. [ILO H,I]	
Education Method	Week 4.1: Guided sessions during which the purpose/possibilities/scope and boundary conditions of the lab will be explained, and the theory components and challenges will be introduced. At the end of this week, students have a draft formulation of the Lab objectives.	

Assessment	Weeks 4.2 4.7: Theory components, in parallel students work on their lab, with bi-weekly review sessions. Compulsory theory units start in week 4.1. All other units start in week 4.2.
	Weeks 4.8 4.10: lab and fieldwork
	This module has two summative assessments:
	1) Labs + theory (80%)
	Group report with individual components + oral presentation.
Expected prior Knowledge	Includes a reflection on the activities related to the individual theory components, as well as an ethical reflection.
	Assessed learning objectives: 1 - 9
	2) Fieldwork (20%)
	Group report + oral presentation
	Assessed learning objectives: 2, 10 - 15
Academic Skills	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
	Atmosphere and Climate Dynamics (discipline core CW) or Module A1, A9, A10
	- Time management
	- Self discipline
	- Ability to read and understand scientific writing
Literature & Study Materials	- Ability to write scientific reports
	- Ability to work in groups, teamwork
	- Critical thinking
	- Computer skills (e.g., programming and scripting)
	- Presentation and public speaking
Judgement	Provided course material, published online lectures and material, book/e-books
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE):
	https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
	There are no exams for this module.
Permitted Materials during Exam	
Collegerama	No

AESM402B	Earth Observation B-module	15
Responsible Instructor	Prof.dr.ing. R. Klees	
Module Manager	Prof.dr.ing. R. Klees	
Instructor	Dr.ir. H. van der Marel	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkoei	
Instructor	Dr. R.C. Lindenbergh	
Instructor	Ir. F.J. van Leijen	
Instructor	Dr.ir. F. Lopez Dekker	
Instructor	Dr. S.L.M. Lhermitte	
Instructor	B.W. van der Wal	
Contact Hours / Week	week 1: 12 PJ	
x/x/x/x	week 3, 5, 6 : 8 PJ	
Education Period	week 7, 8: PJ	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The module consists of four theory units and a lab. Students can choose between two lab categories, i.e., Geohazards and Climate Impacts. The theory units partially depend on this choice. The unit Time Series Analysis is compulsory, at least two units need to be selected from the following units: InSAR data analysis, GNSS Remote Sensing and Positioning, Optical Remote Sensing, Geostatistical Data Analysis (Geohazards lab) / Climate Data Analysis (Climate Impacts lab). The fourth unit can also be chosen from another discipline (see Brightspace). The Lab has a central place in the module, triggering the inquiries and as the place where the students apply the theory to a challenge related to monitoring and/or prediction of climate impacts or geohazards. The student (team, max. 3 students) defines a Lab project related to their chosen challenge. They collect, analyse and interpret data to address or solve their proposed topic. The Lab report includes an individual ethical reflection on the specific challenge and approach. During the fieldwork, students will work in teams and receive the terms of reference as defined by a (virtual) client, with the aim to analyse local to regional deformations and/or mass displacements due to various natural and/or human-induced causes. They need to design a fieldwork campaign accordingly; the design will be reviewed during the project planning review. In the second phase, students collect the data in the field and process historical and newly acquired data. Thereafter they will analyse and interpret the data, present their findings and provide recommendations for future work.	
Study Goals	After completing the module students will be able to: General: 1. Identify open issues in the processing and/or interpretation of Earth system data records based on the outcomes of the lab project and design a development roadmap to address them. 2. Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab and Fieldwork projects in a clear and convincing manner, both orally and written. Theory/Lab (module specific): 3. Assess the value and limitations of Earth observation data in addressing a societal challenge related to geohazards or climate impacts. 4. Select Earth observation data suitable for the study of a phenomenon of interest. 5. Design, implement and validate a workflow to derive/extract geophysical parameters from Earth observation data. 6. Quantify and assess the input data uncertainties and the quality of the results. 7. Characterize temporal and spatial variations of geophysical parameters using Earth observation data. 8. Analyse, visualize and interpret findings in a clear and convincing manner. Fieldwork: 9. Plan and design a field campaign that is appropriate for the physical process to be measured. 10. Collect data in the field using different measurement techniques. 11. Explain and quantify the error sources associated with the field measurements. 12. Process and analyse the data collected in the field to give meaningful constraints on the physical process. 13. Effectively communicate with peers, assessors and clients. 14. Contribute to a project as a team player and to the overall project management.	
Education Method	In week 4.1 students will be informed about the purpose/possibilities/scope as well as the boundary conditions to be met in formulating the objectives of the lab. Moreover, the theory units and associated activities will be introduced. Students make a draft proposal for a Lab project in the first week and will receive feedback on this draft. Moreover, students need to select the theory units. In week 4.3 they will have to submit the final Lab proposal to be approved by the supervisory team. Students will work on the Lab project from week 4.4 onwards. During that period, lab sessions will be scheduled with the supervisor(s) to review and discuss choices, progress, results. Preparation for fieldwork start in week 4.8, actual fieldwork takes place in weeks 4.9 and 4.10.	
Assessment	Formative assessment is based on exercises for the individual theory units, as well as feedback during the lab project by the supervisors. The theory units and the lab project are summatively assessed based on a written report and oral discussion. The report is to be handed in at the end of week 4.7. Report needs to include an individual ethical reflection and an individual accountability report (i.e., describing what the individual contributions were). At the end of week 4.8 there will be an oral discussion with the supervisors. The fieldwork is formatively assessed at the midterm; students will receive individual feedback from staff as well as peer feedback from their team members. Summative assessment is based on technical performance as well as project skills (rubric available on Brightspace). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	AESM3002 combined with one of the A-modules AESM302A, AESM303A, AESM307A, AESM309A; or AESM308A (which includes AESM3002)	
Academic Skills	Data analysis, interpretation, visualization	

Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.
Judgement	n/a
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	n/a
Collegerama	No

AESM403B	Geo-Energy B-module	15
Responsible Instructor	Prof.dr. P.L.J. Zitha	
Module Manager	Prof.dr. P.L.J. Zitha	
Instructor	Dr.ir. G.G. Drijkoningen	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr. J.E.A. Storms	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. P.G. Ditmar	
Instructor	A. Barnhoorn	
Instructor	Dr. G. Bertotti	
Instructor	Prof.dr. L.G. Evers	
Instructor	Dr. A.W. Martinius	
Instructor	Dr. P.B.R. Bruna	
Instructor	Dr. H.A. Abels	
Instructor	Dr. A.M.H. Pluymakers	
Instructor	Dr. A.G. Muntendam-Bos	
Contact Hours / Week x/x/x/x	week 1: 12 PJ week 3, 5, 6 : 8 PJ week 7, 8: PJ	
Education Period	4	
Start Education	4	
Course Language	English	
Course Contents	The lab and the theory part of the B-module are connected and inquiry-based. Theory (8EC) The module includes the following theory components: Compulsory component (2 EC): Induced Seismicity Subset 1 components (at least 4 EC): Geophysical prospecting (2EC), Geological interpretation of seismic data (2EC), Production science and engineering(2EC). Subset 2 components (at most 2 EC): Geostatistical data analysis, Exploration tools and methods, Advanced resources modeling, Time series analysis, (In)SAR, Optical remote sensing, Climate data analysis, Sea level change extremes A short description of each theory component follows: Geophysical prospecting: · Formulate the conceptual relationships between physical properties of the subsurface and potential-, diffusive-, and wave-field measurements. · Explain how these properties can be retrieved through processing or inversion of the collected data. · Choose and apply the proper method(s) to solve a specific problem related to Energy Transition or Geohazard applications. Geological interpretation of seismic data: · Explain what information is present in geophysical data (images and attributes) and to what extent the data can be reliably used for their geological interpretation and apply this to the geophysical data sets. · Provide a geological interpretation of the structure, the sedimentary environment, and the properties of the rocks imaged in a geophysical data set related to Energy Transition and Geohazard applications. Production science and engineering: · Explain the differences between production technology in hydrocarbon, geothermal and subsurface-storage industry, and evaluate risks and safety issues of its well drilling and operations. · Evaluate production processes and facilities in terms of production scenarios and link them to the effects that engineering applications in the subsurface. Induced seismicity: · Describe and explain the mechanisms of, and the effects that engineering applications in the subsurface have on the natural and build environment and society. The focus will be on induced seismicity and subsidence, and cap rock integrity will be briefly discussed. · Integrate knowledge of ethical issues, international standards and risk management systems into Energy Transition and Geohazards applications. Lab (4EC) Labs are conceived as mini-projects which will be done by groups of 2-3 students. Labs play central role in the module. They trigger the questions on the energy transition challenge or the related geohazards that will be addressed by applying the theory components. The student team can define the topic of their Lab within a predefined energy transition challenge or geohazard theme. Examples of themes include: · Gas depletion from the field · Geological hydrogen or carbon dioxide storage · Extraction of geothermal energy The students collect, analyze and interpret data to address or solve their proposed topic. The Lab includes an ethical reflection (0.5 EC) on the specific challenge. Fieldwork (3EC) Group size: 2-3	

Study Goals	Part 1 (0.5EC): preparation Prepare for the assignment in a team. Discussion of the main scientific and technical challenges related to energy transition and/or geohazards. Read literature on the geological objects investigated. Define the data needed and the strategy to collect them, preparing the necessary (digital) platforms. Part 2 (1.5EC): the field trip In the various outcrops visited, student groups will collect the data necessary to address the questions defined in Part 1. Preliminary data processing will be performed during the late afternoon/evening together with the definition of activities needed for the coming day. Part 3 (1EC): reporting The 3rd part is dedicated to the final processing of the acquired data and the preparation of a schematic report on the first-order questions posed in part 2. General learning objectives for theory, labs and fieldwork.
	After completing the module students will be able to: General: Identify open issues in the processing and/or interpretation of Earth system data records based on the outcomes of the lab project and design a development roadmap to address them. [ILO E,F] Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab and Fieldwork projects in a clear and convincing manner, both orally and written. [ILO F,G,H] Energy Transition and Geohazards Lab + Theory Define and solve a research topic related to the Energy Transition or Geohazards challenge. [ILO E] Analyse Earth system processes through a combination of data, observations and model outputs (geophysical data, subsurface data, petrophysical data, and monitoring data). [ILO A,B] Extract subsurface characteristics and evaluate the options and limitations of data types for present-day and future societal challenges in Energy Transition or Geohazards. [ILO C,D] Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab project in a clearly written and convincing manner. [ILO F,G] Fieldwork: Design and execute a fieldwork campaign appropriate for the Earth system processes and/or applications to be studied. [ILO B,C,H,J] Describe, identify and measure sedimentary heterogeneities, faults, fractures, folds, and 2D rock property trends at different scales (10-1 to 103m) and qualitatively predict their architecture and their potential impact on dynamic behaviour. [ILO A,B,C] Link field observations of rock property trends at different sub-seismic scales to geophysical data and include the impact of scale to model uncertainty. [ILO B,C] Extract subsurface characteristics and evaluate the options and limitations of data types for present-day and future societal challenges in Energy Transition and Geohazards. [ILO C,D] Present analyses, interpretations and conclusions of the Fieldlab project in a clearly written and convincing manner. [ILO G] Contribute to a project as a team player and to the overall project management. [ILO H,J]
Education Method	Week 1: guided sessions during which the purpose/possibilities/scope as well as the boundary conditions to be met in formulating the objectives of the lab will be explained, and also the theory components and challenges will be introduced. At the end of this week, students have a draft formulation of the questions they want to answer in the lab.
Assessment	Week 2-7: Theory components, in parallel students work on their lab, with bi-weekly review sessions. Weeks 8-10: lab and fieldwork. This module has two summative assessments: 1) Labs + theory (80%) Individual report. Includes a reflection on the activities related to the individual theory components, as well as an ethical reflection. Assessed learning objectives: 1- 6 2) Fieldwork (20%) Assessed learning objectives: 2, 7-12 3) Formative assignments for the theory components in the form of homework, reporting and/or presentations. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Expected prior Knowledge Academic Skills Geo-Energy Engineering Applications (DC-GE) critical thinking, oral presentation skills, written reporting skills

Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	NA
Collegerama	No

AESM404B	Geo-resources B-module	15
Responsible Instructor	Dr. M.W.N. Buxton	
Module Manager	Dr. M.W.N. Buxton	
Instructor	Dr.ir. G.G. Drijkoningen	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. M. Soleymani Shishvan	
Instructor	Dr. M.A. Schleiss	
Contact Hours / Week x/x/x/x	week 1: 12 PJ week 3, 5, 6 : 8 PJ week 7, 8: PJ	
Education Period	4	
Start Education	4	
Course Language	English	
Course Contents	The lab and the theory part of the B-module are connected and inquiry-based.	
	Theory (8EC)	
	[add text]	
	Lab (4EC)	
	Individual or groups of max. 3 students	
	The lab has a central place in the module, triggering the inquiries and as the place where the students apply the theory to a challenge related to characterisation and modelling of mineral resources in order to maximise resource efficiency and/or resource a green future. The student (team) defines a topic related to the challenge within a predefined theme that they will work on during the Lab. They collect, analyse, model and interpret data to address or solve their posed topic. The lab includes an ethical reflection (0.5 EC) on the specific challenge and approach.	
	Fieldwork (3EC)	
	Group size: 3-5	
	Phase 1: project planning (0.5EC)	
	Teams receive the terms of reference as defined by a (virtual) client, with the aim to analyse specific topic related to resource efficiency and/or resourcing a green future. They need to design a fieldwork campaign accordingly; the design will be reviewed during the project planning review.	
	Phase 2: data collection & processing (1.5EC)	
	Data collection in field and processing of historical and newly acquired data.	
	Phase 3: analysis & interpretation (1EC)	
	Data analysis and interpretation, answering the questions from the terms of reference, reporting in form of a final group presentation and a written report.	
Study Goals	After completing the module students will be able to:	
	General:	
	Identify open issues in the processing and/or interpretation of Earth system data records based on the outcomes of the lab project and design a development roadmap to address them. [ILO E,F]	
	Analyse, visualize and interpret findings in a clear and convincing manner. [ILO B,F]	
	Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab and Fieldwork projects in a clear and convincing manner, both orally and written. [ILO F,G,H]	
	Theory/Lab (module specific):	
	Assess the value and limitations of mineral exploration data in addressing a societal challenge related to resource efficiency and responsible sourcing of mineral resources for a green future. [ILO A,B,E]	
	Select appropriate exploration tools and methods and design a workflow to acquire the necessary data for mineral resource characterisation and modelling in order to select a suitable extraction method. [ILO A,B]	
	Analyse and manipulate data to assess the degree of spatial continuity and use deterministic and/or geostatistical interpolation methods to estimate unknown variables of interest. [ILO B]	
	Model the data using advanced geostatistical techniques to define a quantified mineral resource incorporating an estimation of the associated degree of confidence and uncertainty in compliance with international reporting standards (e.g. JORC, PERC). [ILO B,C]	
	Fieldwork:	
	plan a field campaign that is appropriate for a selected deposit style and environmental impact to be measured. [ILO A,D,H,J]	
	collect data in the field using different measurement tools and techniques. [ILO A,J]	
	explain and quantify the error sources and uncertainties associated with the field measurements. [ILO A,B]	
	identify and justify the choice of the selection of extraction methods. [ILO A,B]	
	process and analyse the data collected in the field to assess possible environmental consequences. [ILO B,C]	

	effectively communicate with peers, assessors and clients.[ILO G,H,I]
	contribute to a project as a team player and to the overall project management. [ILO H,J]
Education Method	Week 1: guided sessions during which the purpose/possibilities/scope as well as the boundary conditions to be met in formulating the objectives of the lab will be explained, and also the theory components and challenges will be introduced. At the end of this week, students have a draft formulation of the questions they want to answer in the lab. Week 2 7: Theory components, in parallel students work on their lab, with bi-weekly review sessions. Weeks 8 10: lab and fieldwork.
Assessment	This module has two summative assessments: 1) Labs + theory (80%) Individual report. Includes a reflection on the activities related to the individual theory components, as well as an ethical reflection. Assessed learning objectives: 1-7 2) Fieldwork (20%) Project including a group presentation and a written report. Assessed learning objectives: 3, 8-14 A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	DC-GR
Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

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